Date

电电复习

一. 电路定律

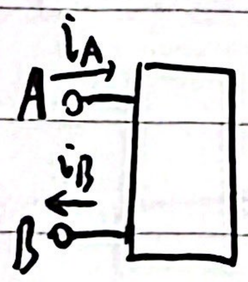
1. $i(t) = I_{DC} + i_{ac}(t)$

$i_{ac}(t) = i(t) - \overline{i(t)}$ $\quad \overline{i_{ac}(t)} = \overline{i(t) - \overline{i(t)}} = 0$

2. Kirchhoff Current Law : KCL

KVL

二. 电阻与电源

1. 

端口条件：$i_A = i_B$

准静态条件：$d_{AB} \ll \lambda$

2. 有源 / 无源

3. 线性 / 非线性

$v(t) = f(i(t))$ $\quad$ 线性 $\Leftrightarrow f(\alpha i_1 + \beta i_2) = \alpha f(i_1) + \beta f(i_2) = \alpha v_1 + \beta v_2$

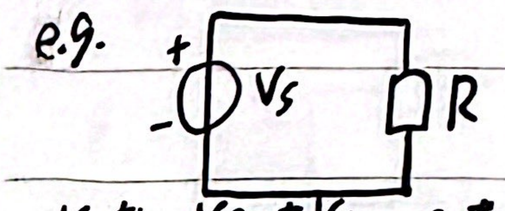
4. 时变 / 时不变

"不同时刻加激励，v-i 关系（图象）变不变"

（开关问题：见 ppt）

5. 有效值

"平方平均开根号"

e.g. 

$v_S(t) = V_{S0} + V_{Sp}\cos\omega t$

$P_L(t) = \frac{v_L^2(t)}{R} = \frac{(V_{S0} + V_{Sp}\cos\omega t)^2}{R} = \frac{1}{R}\left[V_{S0}^2 + 2V_{S0}V_{Sp}\cos\omega t + V_{Sp}^2\frac{(1+\cos 2\omega t)}{2}\right]$

$P_L = \overline{P_L(t)} = \frac{V_{S0}^2 + \frac{1}{2}V_{Sp}^2}{R} = P_{DC} + P_{AC}$

6. （课后） 任意一个信号可分为直流信号分量与交流信号分量之和

$f(t) = f_{DC} + f_{AC}(t)$

$f_{DC} = \overline{f(t)} = \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f$ $\quad$ （计 $f(t+T) = f(t)$，不必取 lim）

例：

该信号 $f_{DC} = 0$ 为纯交流信号

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三. 电路定理

1. 对偶原理

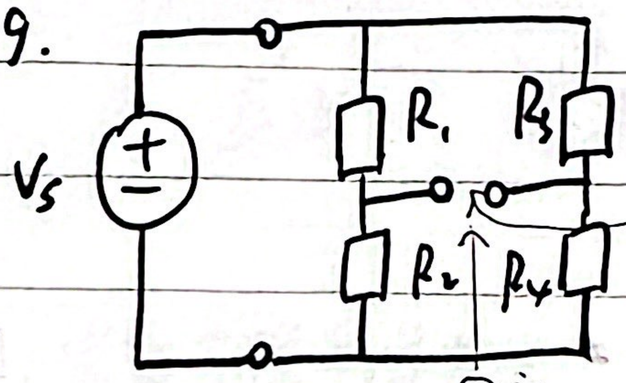
2. 单端口等效　　注：惠斯通电桥

① 加压求流 (流) (压)
- 硬算
- 结点电压法
- 回路电流法

② 用电路定理对一些简单的常见模型 快速给出等效

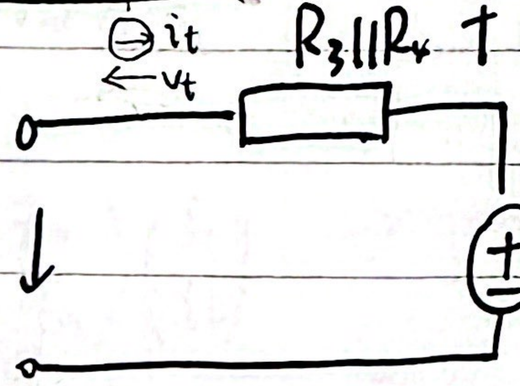
(注：先②简化一部分再①是常用方法)

e.g.



求等效电压源时，端口开路

求等效内阻时，内部源不起作用（独立）

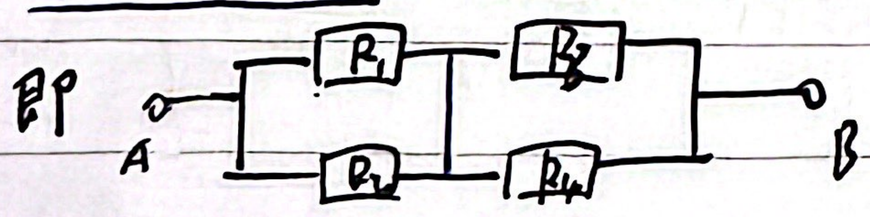


⇒ 等效为：

这与所谓的戴维南定理是一样的：

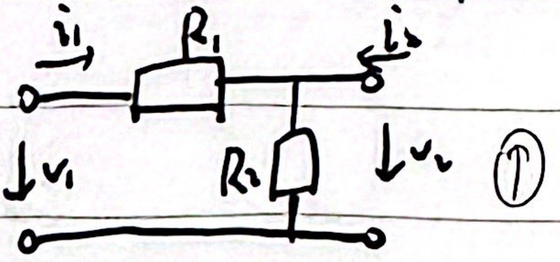
① 戴维南电压：端口开路电压

② 戴维南内阻：内部独立源不起作用时端口看入电阻



3. 二端口等效 (6种方案)

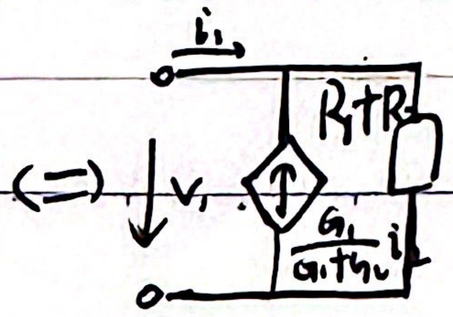
例：

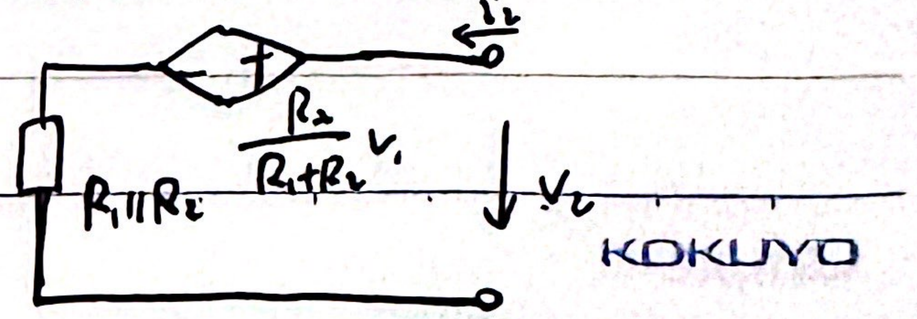


计1加V，2加i

则以 V_1、i_2 作自变量，V_2、i_1 作因变量，获得如下等效：

$$\Rightarrow \begin{cases} i_1 = \frac{1}{R_1+R_2} V_1 - \frac{G_1}{G_1+G_2} i_2 \\ V_2 = \frac{R_2}{R_1+R_2} V_1 + \frac{1}{G_1+G_2} i_2 \end{cases}$$





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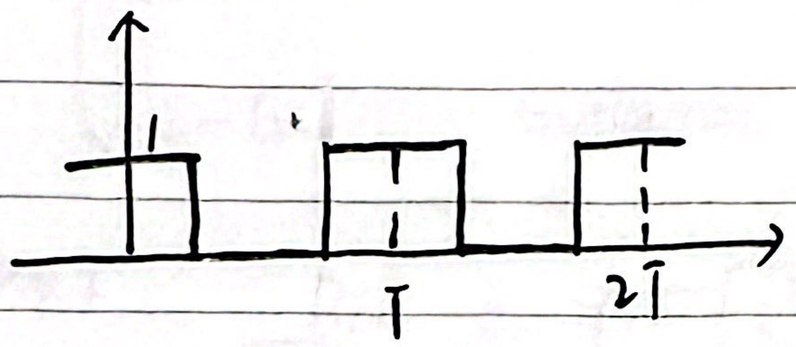
以受控源为核心的四种理向放大器

VCVS	VCCS	CCCS	CCVS
电压	互导	电流	互阻

4. 方波信号的 Fourier 展开



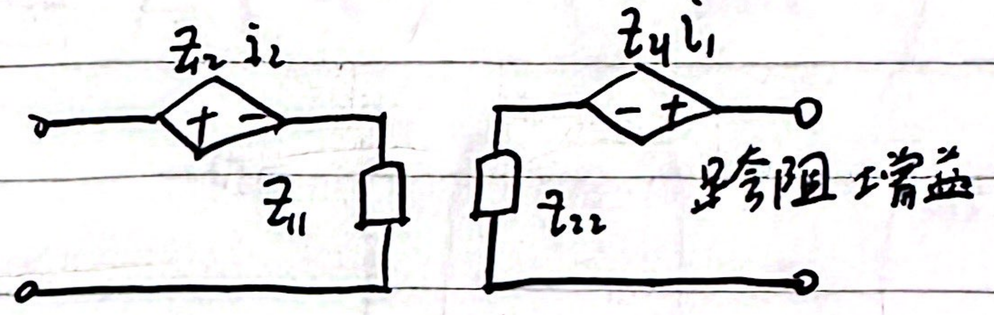
$$S_1(t) = \frac{1}{2} + \frac{2}{\pi}\cos\omega_0 t - \frac{2}{3\pi}\cos 3\omega_0 t + \frac{2}{5\pi}\cos 5\omega_0 t - \cdots$$

直流　基波　三次谐波　五次谐波

四. 完备方程列写方法

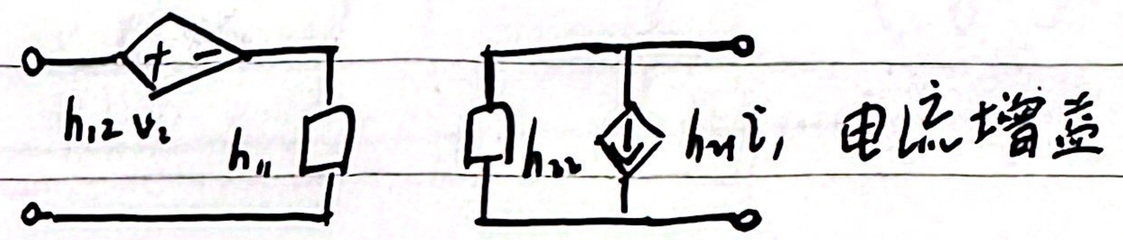
1. 网络参量

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$$



...

$$\begin{pmatrix} V_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ V_2 \end{pmatrix}$$



...

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_2 \\ -I_2 \end{pmatrix}$$ 传输参量矩阵

...

$$\begin{cases} A = \left.\frac{V_1}{V_2}\right|_{I_2=0} = \frac{1}{g_{21}} \\ B = -\left.\frac{V_1}{I_2}\right|_{V_2=0} = \frac{1}{-y_{21}} \\ \cdots \end{cases}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} A & -B \\ -C & D \end{pmatrix}^{-1}$$

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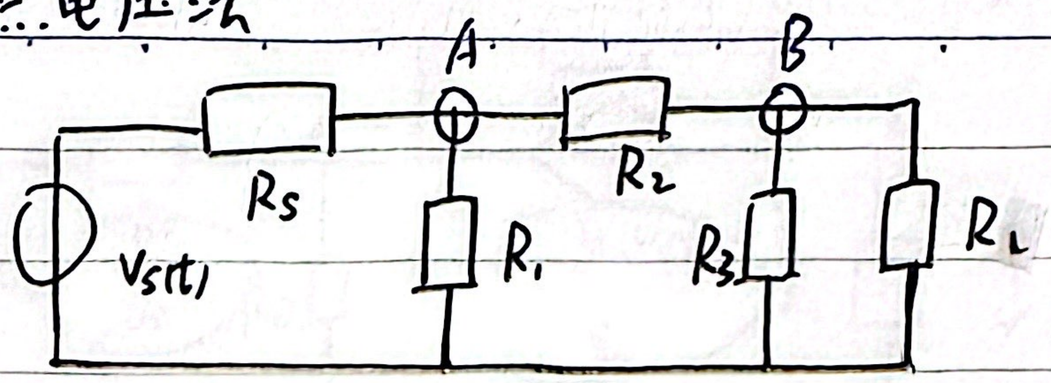
2. 结点电压法

e.g.1



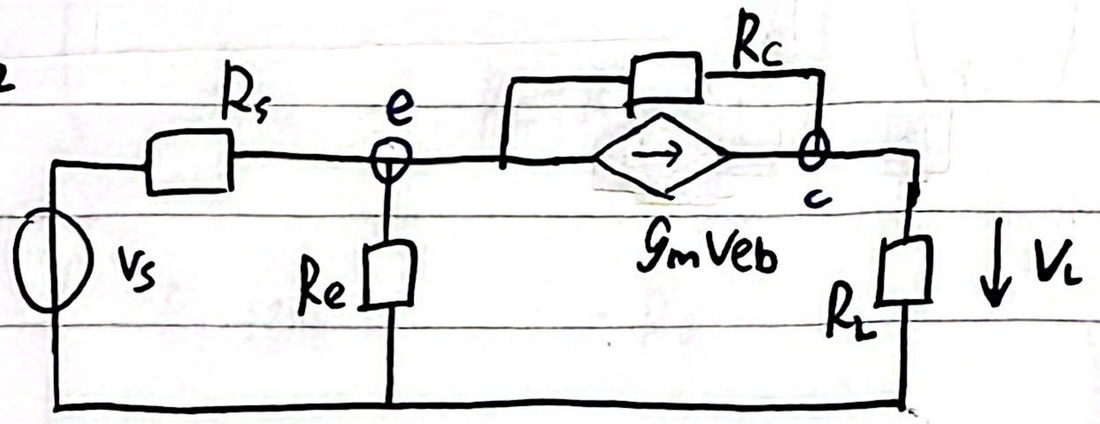
(戴源→诺源)

$$\begin{pmatrix} G_S+G_1+G_2 & -G_2 \\ -G_2 & G_2+G_3+G_L \end{pmatrix}\begin{pmatrix} V_A \\ V_B \end{pmatrix} = \begin{pmatrix} G_S V_S \\ 0 \end{pmatrix}$$

↗流入结点电流(-流出)

11
0

e.g.2



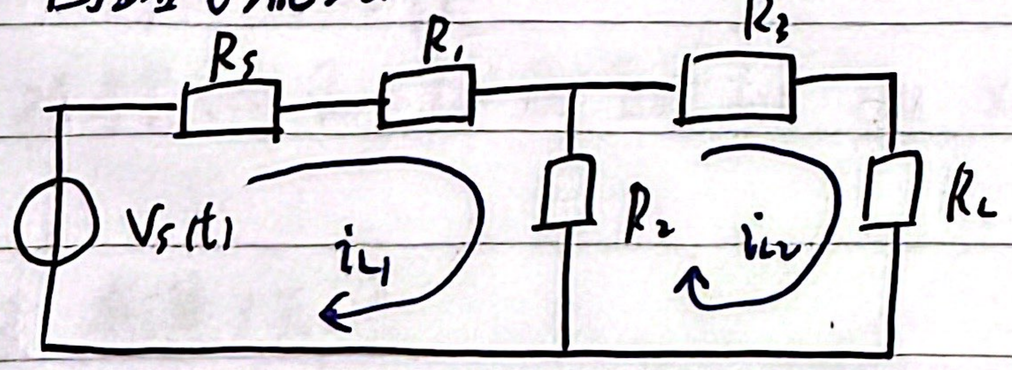
流入-流出

$$\begin{pmatrix} G_S+G_e+G_C & -G_C \\ -G_C & G_C+G_L \end{pmatrix}\begin{pmatrix} V_e \\ V_C \end{pmatrix} = \begin{pmatrix} G_S V_S - g_m V_e \\ g_m V_e \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} G_S+G_e+G_C+g_m & -G_C \\ -g_m-G_C & G_C+G_L \end{pmatrix}\begin{pmatrix} V_e \\ V_C \end{pmatrix} = \begin{pmatrix} G_S V_S \\ 0 \end{pmatrix}$$

写答案要写成这样!

3. 回路电流法



$$\begin{pmatrix} R_S+R_1+R_2 & -R_L \\ -R_L & R_2+R_3+R_L \end{pmatrix}\begin{pmatrix} i_{L1} \\ i_{L2} \end{pmatrix} = \begin{pmatrix} V_S \\ 0 \end{pmatrix}$$

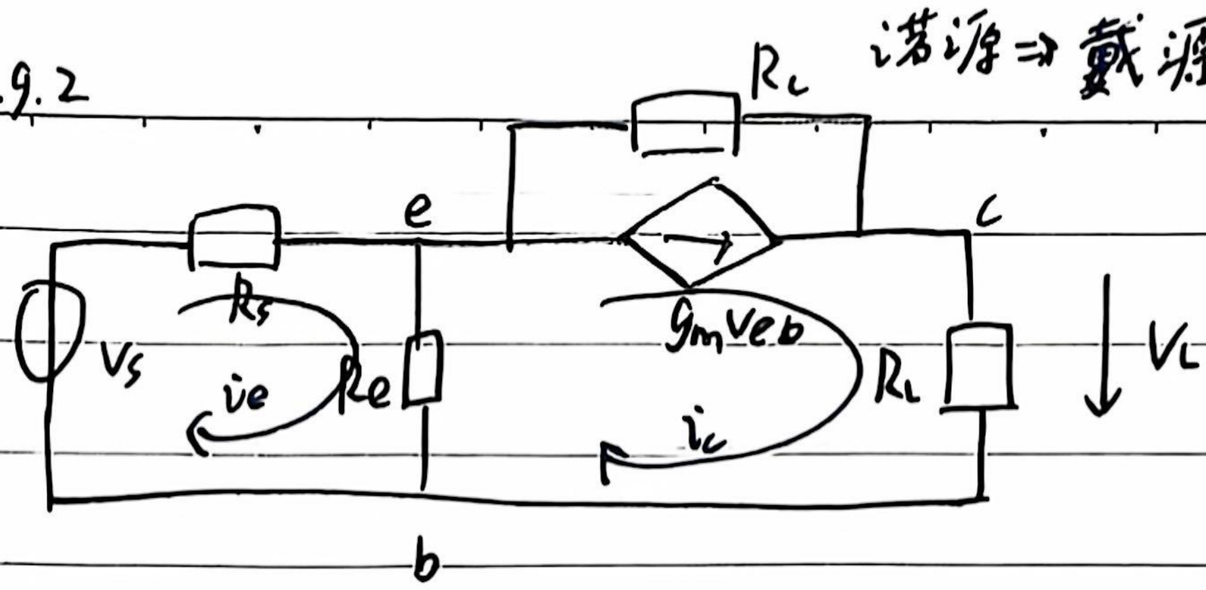
i_{L1} 扫过的电阻

i_{L1}、i_{L2} 共同扫过的电阻，电流反向扫∴-

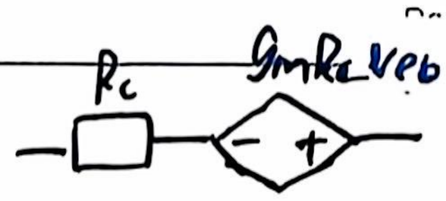
→ 回路中的电动势

(从-扫到+ ∴写+号)

e.g.2



诺源⇒戴源



$$\begin{pmatrix} R_s+R_e & -R_e \\ -R_e & R_c+R_L+R_e \end{pmatrix}\begin{pmatrix} i_e \\ i_c \end{pmatrix}=\begin{pmatrix} V_s \\ g_mR_cV_e \end{pmatrix}$$

$\rightarrow V_e=R_e(i_e-i_c)$

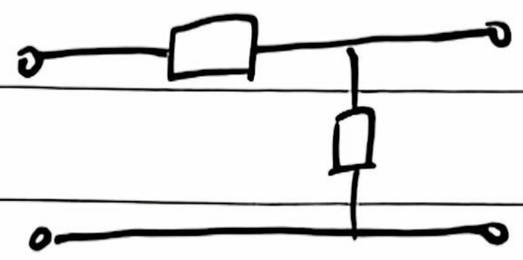
$$\Rightarrow \begin{pmatrix} R_s+R_e & -R_e \\ -R_e-g_mR_cR_e & g_mR_cR_e+R_c+R_L+R_e \end{pmatrix}\begin{pmatrix} i_e \\ i_c \end{pmatrix}=\begin{pmatrix} V_s \\ 0 \end{pmatrix}$$

五. 网络分析

1. 互易 / 非互易

互易Thm: 互易 $\Leftrightarrow z_{12}=z_{21} \Leftrightarrow y_{12}=y_{21} \Leftrightarrow h_{12}=-h_{21} \Leftrightarrow g_{12}=-g_{21}$

e.g.



由互易元件组成的网络一定是互易网络

2. 对称 / 非对称

对称 $\Leftrightarrow z_{11}=z_{22}$ 且 $z_{12}=z_{21} \Leftrightarrow y_{11}=y_{22}$ 且 $y_{12}=y_{21} \Leftrightarrow \Delta_h=1$ 且 $h_{12}=-h_{21}$

$\Leftrightarrow \Delta_T=1$ 且 $A=D$

3. 有源 / 无源

$\Leftrightarrow$ 端口吸收除恒 ≥ 0

设 P 可表示 Z Y G H $\quad P=\begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix}$

有源 $\Leftrightarrow$ ① $P_{11}<0$ ② $P_{22}<0$ (负阻有源性)

③ $(P_{12}+P_{21})^2-4P_{11}P_{22}>0$ (受控源有源性)

(3选一即有源)

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4. 针对无源网络定义 无损 / 有损

端口吸收总功率恒 = 0 ⟺ $P_{11}=0$ 且 $P_{22}=0$ 且 $P_{12}=-P_{21}$

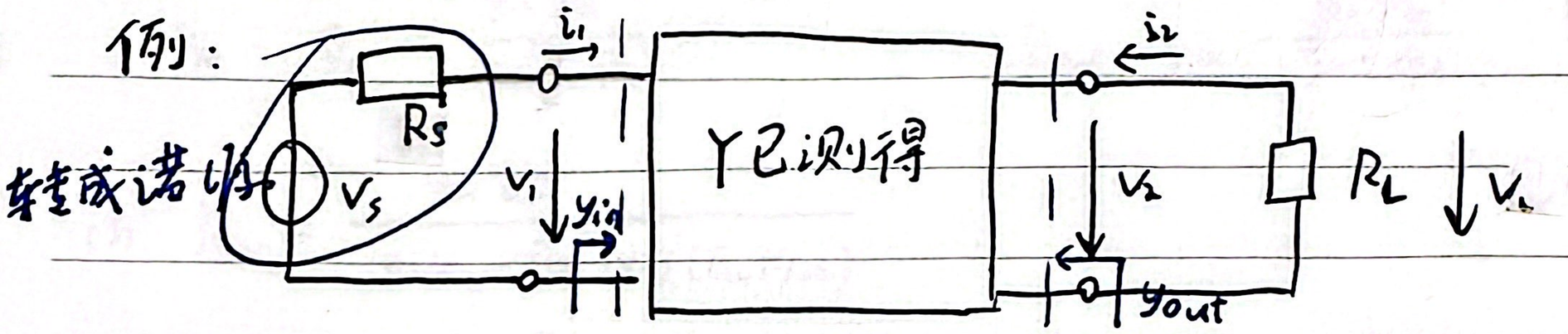
5. 单向 / 双向 ⟺ $P_{12}\neq 0$ 且 $P_{21}\neq 0$

P_{12}, P_{21} 中有一个 = 0 ⟺ $\Delta r = 0$ ($\Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ 奇)

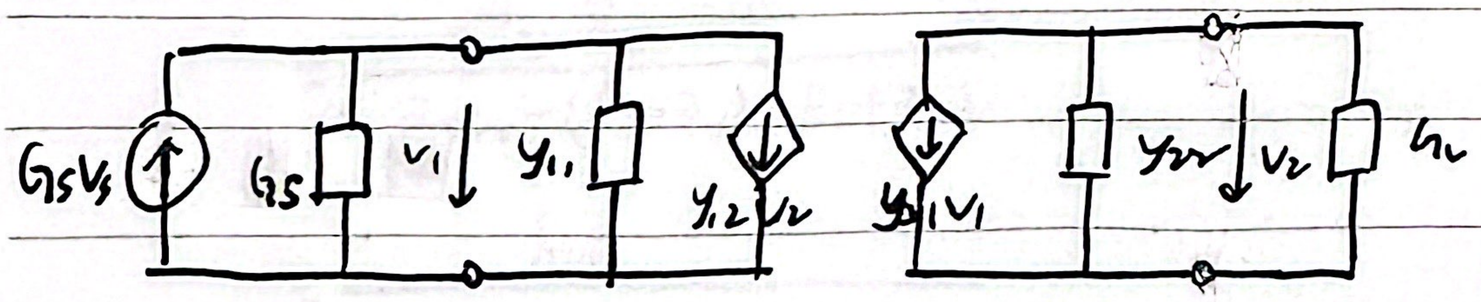
6. 求传递函数

(假设网络参量已测得. how 求传函?)

例:



转成诺顿

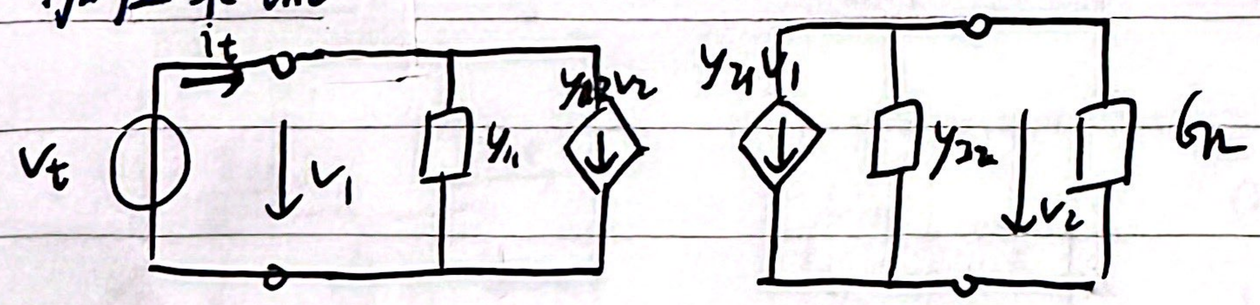


$$V_2 = \frac{-y_{21}V_1}{y_{22}+G_L} = \frac{-y_{21}}{y_{22}+G_L}\cdot\frac{G_S V_S - y_{12}V_2}{G_S+y_{11}}$$

$$\Rightarrow H = \frac{V_L}{V_S} = \frac{V_2}{V_S} = \frac{y_{21}G_S}{y_{21}y_{12}-(G_S+y_{11})(G_L+y_{22})}$$

求 y_{in}, y_{out}

加压求流



$$i_t = V_1 y_{11} + y_{12}V_2 = V_t y_{11} - y_{12}\frac{y_{21}V_t}{y_{22}+G_L}$$

$$\Rightarrow y_{in} = y_{11} - \frac{y_{12}y_{21}}{y_{22}+G_L}$$

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结论：

(1) $y_{in} = y_{11} - \frac{y_{12}y_{21}}{y_{22}+G_L}$ $\quad y_{out} = y_{22} - \frac{y_{12}y_{21}}{y_{11}+G_S}$

$z_{in} = z_{11} - \frac{z_{12}z_{21}}{z_{22}+R_L}$ $\quad z_{out} = z_{22} - \frac{z_{12}z_{21}}{z_{11}+R_S}$

(2) $z_{in} = h_{11} - \frac{h_{12}h_{21}}{h_{22}+G_L}$ $\quad y_{out} = h_{22} - \frac{h_{12}h_{21}}{h_{11}+R_S}$

用量纲记！

$y_{in} = g_{11} - \frac{g_{12}g_{21}}{g_{22}+R_L}$ $\quad z_{out} = g_{22} - \frac{g_{12}g_{21}}{g_{11}+G_S}$

(注 $\begin{pmatrix} i_1 \\ v_2 \end{pmatrix} = G\begin{pmatrix} v_1 \\ i_2 \end{pmatrix}$ 不还是上172)

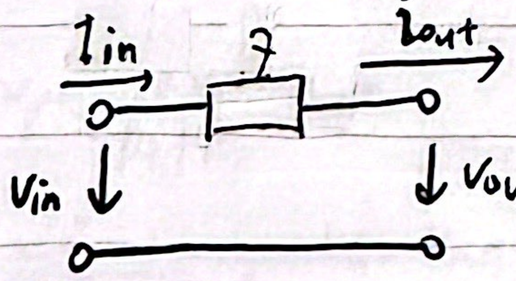
(3) $|H_V| = \frac{y_{21}G_S}{y_{12}y_{21} - (G_S+y_{11})(G_L+y_{22})}$

$= \frac{-z_{21}R_L}{z_{12}z_{21} - (R_S+z_{11})(R_L+z_{22})}$

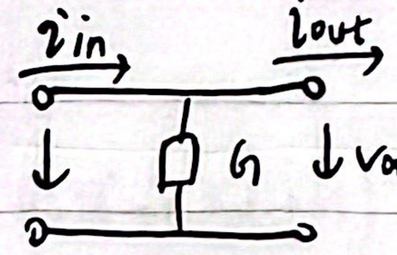
$= \frac{h_{21}}{h_{12}h_{21} - (R_S+h_{11})(G_L+h_{22})}$

$= \frac{-g_{21}G_SR_L}{g_{12}g_{21} - (G_S+G_{11})(R_L+g_{22})}$

7.

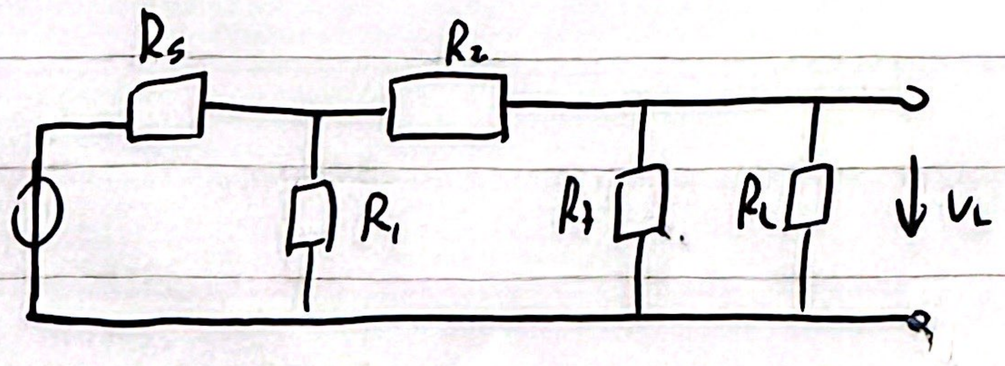


$\begin{pmatrix} v_{in} \\ i_{in} \end{pmatrix} = \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix}\begin{pmatrix} v_{out} \\ i_{out} \end{pmatrix}$



$\begin{pmatrix} v_{in} \\ i_{in} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix}\begin{pmatrix} v_{out} \\ i_{out} \end{pmatrix}$

例：



$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & R_S \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ G_1 & 1 \end{pmatrix}\begin{pmatrix} 1 & R_2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ G_3+G_L & 1 \end{pmatrix}$

$|H_V| = \frac{v_L}{v_S} = \frac{1}{A}$.

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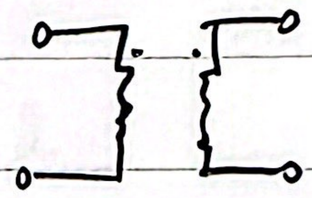
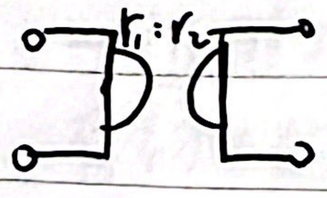
8. 理想变压器

$$\begin{pmatrix} v_1 \\ i_1 \end{pmatrix} = \begin{pmatrix} n & 0 \\ 0 & \frac{1}{n} \end{pmatrix} \begin{pmatrix} v_2 \\ -i_2 \end{pmatrix}$$



9. 理想回旋器

$r_1 = r_2 \Leftrightarrow$

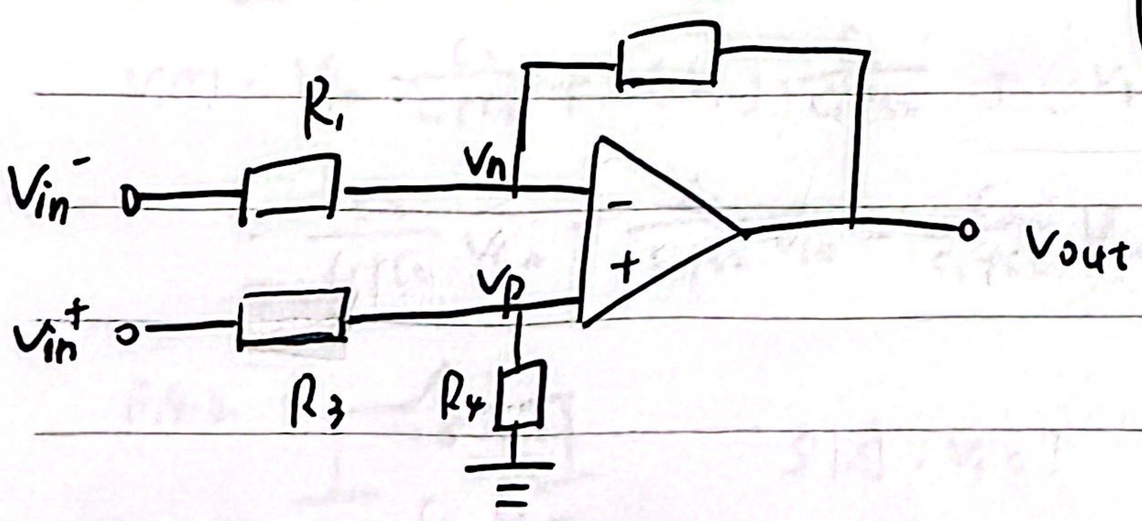


$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 & -r_1 \\ r_2 & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$$

六. 电容电感

1. 运算放大器

e.g. 差分放大电路：



（注：v_{ip}：同相输入端
v_{in}：反相输入端
$v_{id} = v_{ip} - v_{in}$ 差模输入）

$$\begin{cases} v_p = \dfrac{R_4}{R_3 + R_4} v_{in}^+ \\ \dfrac{v_{in}^- - v_n}{R_1} = \dfrac{v_n - v_{out}}{R_2} \\ v_n = v_p \end{cases} \Rightarrow v_{out} \xrightarrow{\text{令 } R_3 = R_1,\ R_4 = R_2} \frac{R_2}{R_1}\left(v_{in}^+ - v_{in}^-\right)$$

（其余见笔记&ppt）

2. C.L

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七. 状态初值的源等效

1. $\delta(t)$ 与 $u(t)$

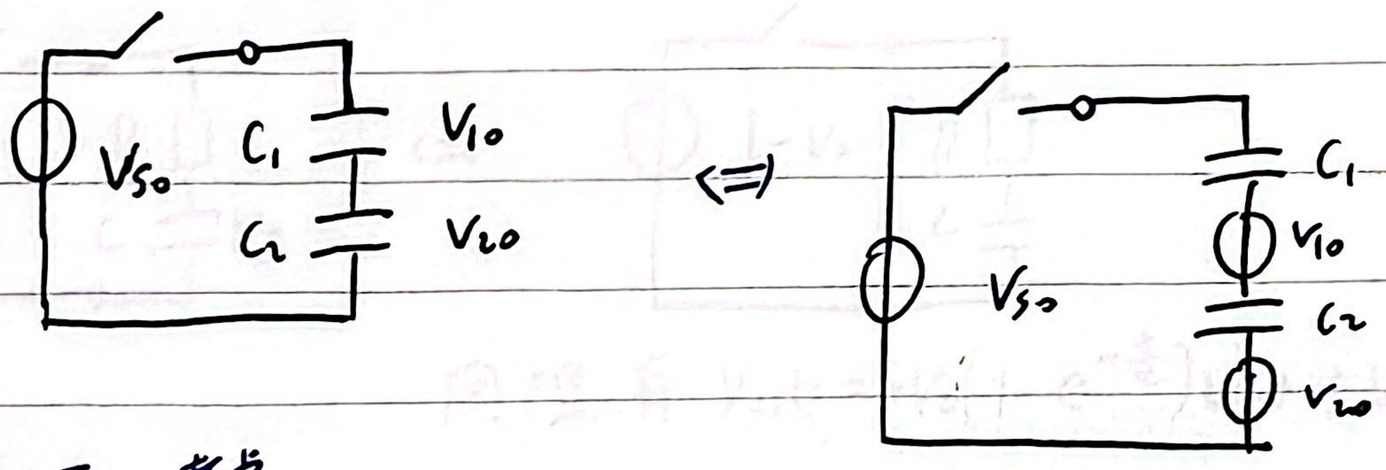
注：当冲激source时，C的V可跳变

2. 有V_0的C的串并联 （与等效）

八. 一阶时域

1. 有初V的C

e.g. 1

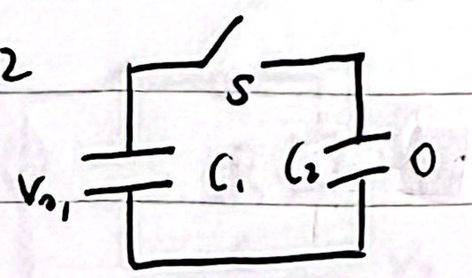


可从叠加Thm考虑：

$$V_1(t) = V_{S0}\cdot\frac{C_2}{C_1+C_2} + (-V_{10})\cdot\frac{C_2}{C_1+C_2} + (-V_{20})\cdot\frac{C_2}{C_1+C_2} + V_{10}$$

$$= \frac{C_2}{C_1+C_2}V_{S0} + \frac{C_1}{C_1+C_2}V_{10} - \frac{C_2}{C_1+C_2}V_{20}$$

e.g. 2

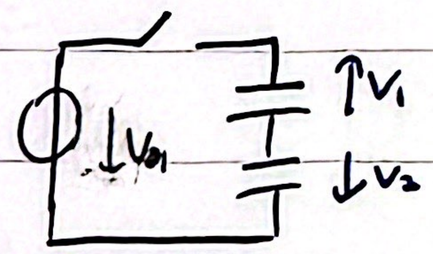


S闭，V_1=?

法一）电荷守恒 $C_1V_{01} = C_1V_1 + C_2V_2 \xrightarrow{V_1=V_2} (C_1+C_2)V_1 \Rightarrow V_1 = \frac{C_1V_{01}}{C_1+C_2}$

法二）等效

⟺

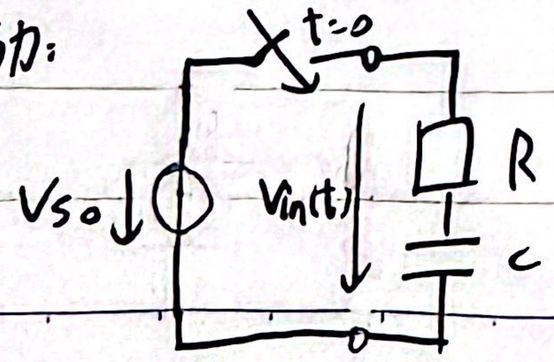


$$V_1 = -\frac{C_2}{C_1+C_2}V_{01} + V_{01} = \frac{C_1}{C_1+C_2}V_{01}$$

2. 阻容分压电路

(1) 直流激励后：C开路

(2) 阶跃激励后：



$$V_{in}(t) = V_{S0}\,u(t)$$

$$= i(t)R + \frac{1}{C}\int_{-\infty}^{t} i$$

$$\Rightarrow \frac{d}{dt}i(t) + \frac{1}{\tau}i(t) = \frac{V_{S0}}{R}\,\delta(t)$$

$$\Rightarrow i(t) = \frac{V_{S0}}{R}e^{-\frac{t}{\tau}}u(t) \quad (\tau = RC)$$

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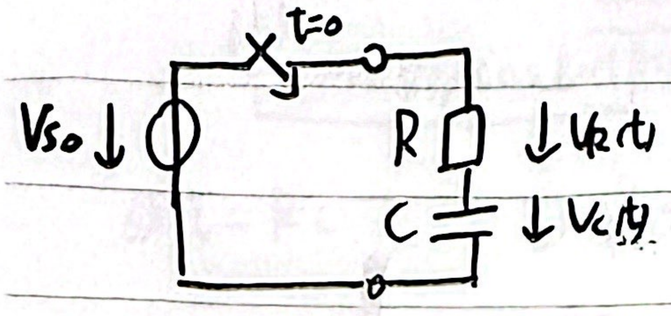
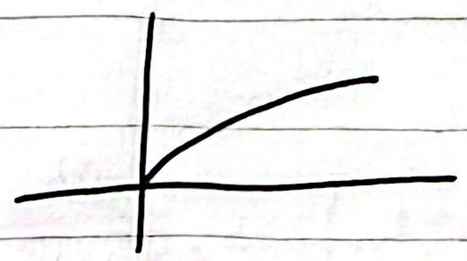
(3) 零状态响应



由(2), $i(t) = \frac{V_{S0}}{R} e^{-\frac{t}{\tau}} U(t)$

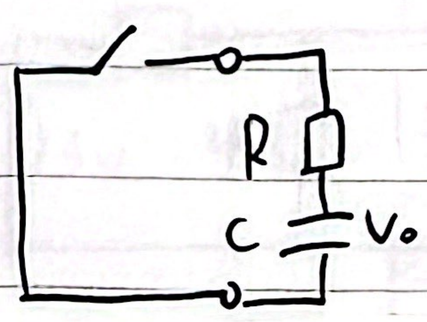
$\Rightarrow \underline{V_C(t)} = V_{S0}(t)U(t) - i(t)R$

$= \underline{V_{S0}\left(1-e^{-\frac{t}{\tau}}\right)U(t)}$

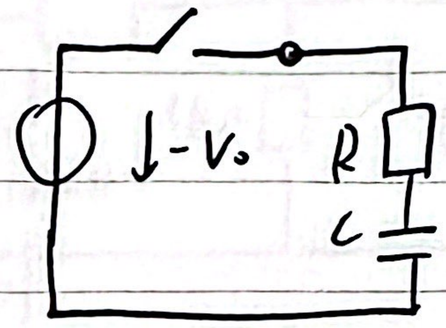


5τ后，认为C充电结束，进入稳态

(4) 零输入响应

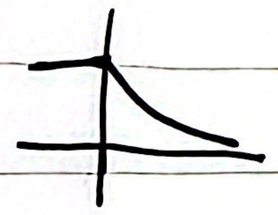


$\Leftrightarrow$



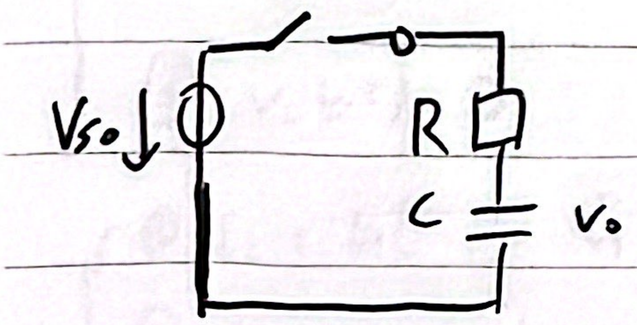
同理，有 $V_C(t) = -V_0\left(1-e^{-\frac{t}{\tau}}\right)U(t) + V_0$

$= \begin{cases} V_0 & t<0 \\ V_0 e^{-\frac{t}{\tau}} U(t) & t>0 \end{cases}$



5τ后，认为C放电结束，进入稳态

(5) 全响应 = 零状态 + 零输入

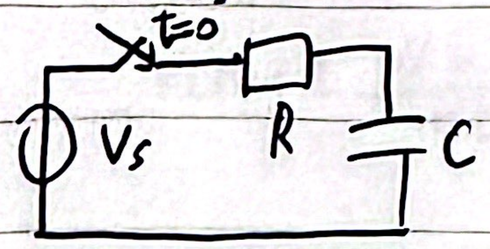


$V_C(t) = (V_{S0} - V_0)\left(1-e^{-\frac{t}{\tau}}\right)U(t) + V_0$

九. 正弦稳态

1. 相量法　略

2. 三要素法



$X(t) = (X_0 - X_{\infty 0})e^{-\frac{t}{\tau}} + X_{\infty}(t)$

要素1：时间常数τ　　$\tau = RC$　　$\tau = GL$

2：初值 X_0

3：稳态响应 $X_\infty(t)$　　$X_\infty(t) = \left(\int_{-\infty}^{t} e^{\frac{\Delta - t}{\tau}} s(\Delta)\, d\Delta\right) U(t)$

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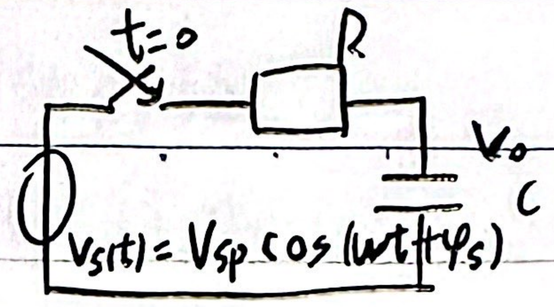
e.g.1

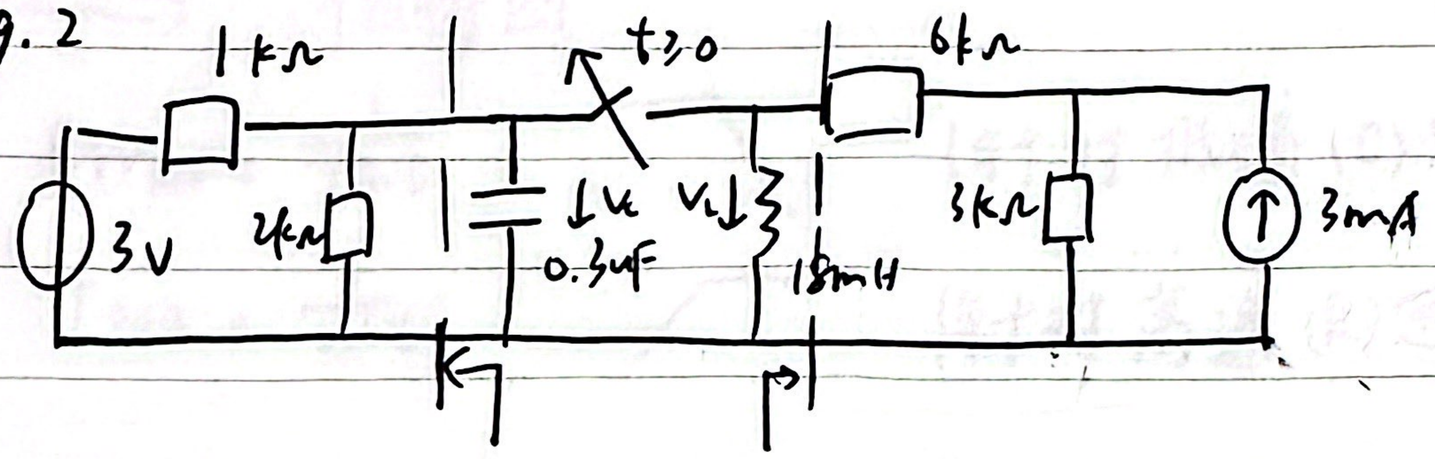


$V_c(t) = (V_{c0^+} - V_{\infty 0})\, e^{-\frac{t}{\tau}} + V_{\infty}(t)$

① $\tau = RC$ ② $V_c(0^+) = V_0$

③ $V_{\infty}(t) = \dfrac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R} V_s(t) = \dfrac{1}{1 + j\omega CR} V_{sp}\cos(\omega t + \varphi_s) = \dfrac{V_{sp}}{\sqrt{1+(\omega\tau)^2}} \cos\left(\omega t + \varphi_s - \arctan\omega\right.$

e.g.2



求断后 $V_c(t)$, $V_L(t)$ (断开前已经)

$V_c(t) = [V_c(0^+) - V_{c\infty}(0^+)]\, e^{-\frac{t}{\tau_c}} + V_{c\infty}(t)$ $(t>0)$

① $\tau_c = RC$ R 等效戴维南 $\frac{1\cdot 2}{1+2} = \frac{2}{3}k\Omega$

② $V_{c\infty}(t) = \frac{2}{2+1}\cdot 3 = 2V$

③ $V_c(0^+) = 0$

① $\tau_L = GL$ $G = \frac{1}{4}mS$

② $i_{L\infty}(t) = 1V$

③ $i_L(0^+) = 3+1 = 4mA$

$\Rightarrow V_C(t) =$

$i_L(t) =$

$V_L(t) = L\dfrac{di_L(t)}{dt} =$

/ 直接求 $V_L(t)$ $V_L(0^+) = -4\times 6 + [-4-(-3)]\cdot 3$

$= -27V$

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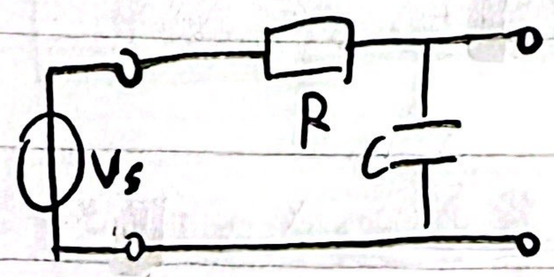
十一 阶时频

1. RC 传函



$$H_{RC}(j\omega)=\frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C}+R}=\frac{1}{\sqrt{1+(\omega RC)^2}}e^{-j\arctan\omega RC}$$

$$=A(\omega)e^{j\varphi(\omega)}$$

记 $\omega_0=\frac{1}{\tau}=\frac{1}{RC}$

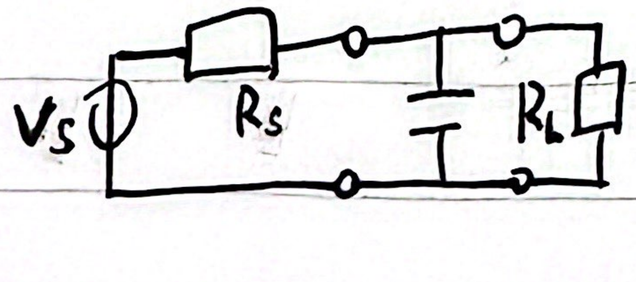
折线近似 → 伯特图

$BW_{3dB}=\frac{1}{2\pi\tau}$ （针对低通(C)定义）

$f_{3dB}=\frac{1}{2\pi\tau}$ （针对高通(R)定义）

·（RL 与之对偶）

一般地，对于一阶低通，传函可写作 $H(j\omega)=H_0\frac{1}{1+j\omega\tau}$



$H_0=H(j0)=\frac{R_L}{R_S+R_L}$

$\tau=RC$ 其中 $R=\frac{R_SR_L}{R_S+R_L}$

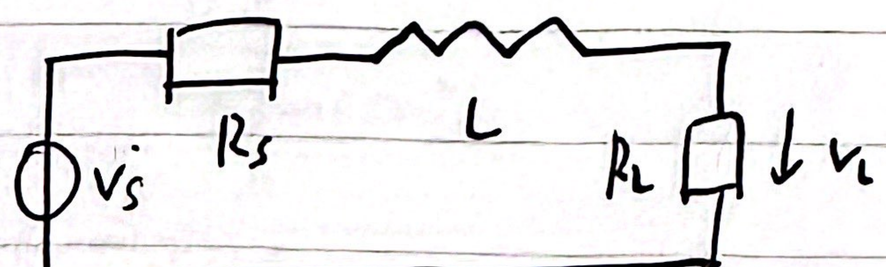
传函的另一种定义方法：

$$H(j\omega)=2\sqrt{\frac{R_S}{R_L}}\frac{\dot V_L(j\omega)}{\dot V_S(j\omega)}$$

2. 阶跃响应与冲激响应

g(t) ↘ 单位 ↙ h(t)

例：



$H(j\omega)=H_0\frac{1}{1+j\omega\tau}$

设 $V_S(t)=V_{S0}U(t)$，求 $H(j\omega)$、BW_{3dB}、$V_L(t)$

$H_0=\frac{R_L}{R_L+R_S}$ $\tau=G_L=\frac{1}{R_L+R_S}L$ $BW_{3dB}=\frac{1}{2\pi\tau}$

$$V_L(t)=\left(V_L(0^+)-V_{L\infty}(0^+)\right)e^{-\frac{t}{\tau}}+V_{L\infty}(t)$$

$V_L(0^+)=0$　$V_{L\infty}(0^+)=\frac{R_L}{R_L+R_S}V_{S0}$　$V_{L\infty}(t)=\frac{R_L}{R_L+R_S}V_{S0}$

阶跃电压不能使状态变量（v_C、i_L）跳变！

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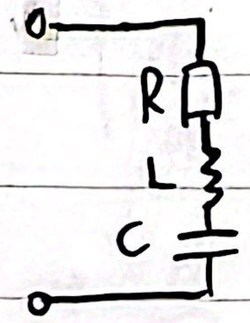
十一. 二阶RLC

1. LC谐振回路

(串联 $\frac{1}{sC}$ ⇔短路，并联 $\frac{1}{sC}$ ⇔开路 ；纯LC谐振无能量损耗)

串联RLC系统参量：



$$\begin{cases} \omega_0 = \frac{1}{\sqrt{LC}} & \text{谐振频率} \\ Z_0 = \sqrt{\frac{L}{C}} & \text{特征阻抗} \\ Q = \frac{Z_0}{R} & \text{品质因数} \quad \left(Q = \frac{\text{电抗}}{\text{电阻}} = \frac{\omega_0 L}{R} = \frac{1/\omega_0 C}{R}\right) \\ \zeta = \frac{1}{2Q} & \text{阻尼系数} \end{cases}$$

$$Z = R + jZ_0\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) = R\left[1 + jQ\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right]$$

对偶地，得到并联RLC

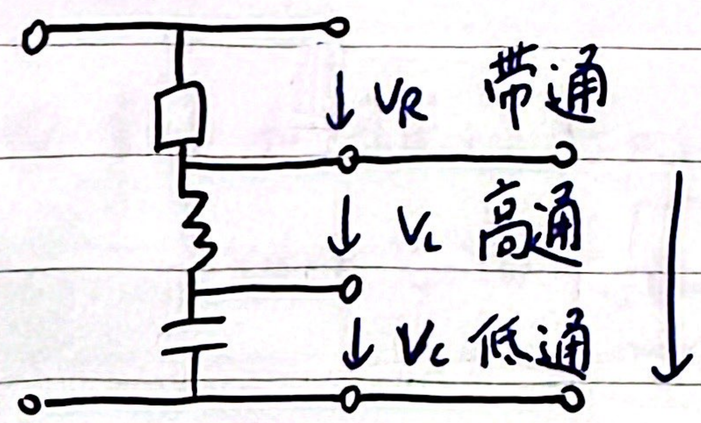
$$\begin{cases} \omega_0 = \frac{1}{\sqrt{LC}} \\ Y_0 = \sqrt{\frac{C}{L}} \\ Q = \frac{Y_0}{G} \\ \zeta = \frac{1}{2Q} \end{cases}$$

注：以上 ω_0 ζ Z_0 Y_0 Q 可背；对于较复杂的电路常用两种方法：

① 频域求 $H(s) = \frac{\cdots}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$

② 时域列方程：$\ddot{v}_0 + 2\zeta\omega_0 \dot{v}_0 + \omega_0^2 v_0 = \cdots$

2. 二阶传函



中心频点 H_0

$H_{VC} = H_0 \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$	0	1	
$H_{VL} = H_0 \frac{s^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$	∞	1	
$H_{VR} = H_0 \frac{2\zeta\omega_0 s}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$	ω_0	1	
$H_{VLC} = H_0 \frac{s^2 + \omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$	0 或 ∞	1	

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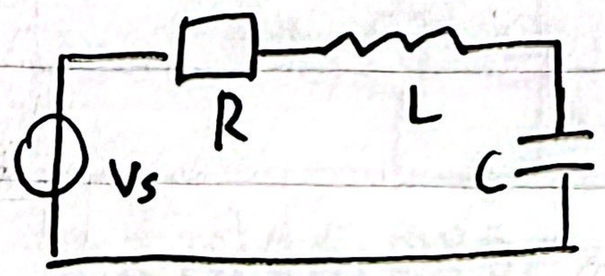
3. RLC串联电路的一般化分析：状态方程法



左侧是状态变量的微分，右侧是状态变量的代数方程

$$\begin{cases} \frac{dv_c}{dt} = \frac{1}{C} i_c = \frac{1}{C} i_L \\ \frac{di_L}{dt} = \frac{1}{L} v_L = \frac{1}{L}(V_S - i_L R - v_c) \end{cases}$$

$$\Rightarrow \frac{d}{dt}\begin{pmatrix} v_c \\ i_L \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{C} \\ -\frac{1}{L} & -\frac{R}{L} \end{pmatrix}\begin{pmatrix} v_c \\ i_L \end{pmatrix} + \begin{pmatrix} 0 \\ V_S/L \end{pmatrix}$$

4. 五要素法

① $X(t) = X_\infty(t) + (X_0 - X_{\infty 0}) e^{-\zeta\omega_0 t} \cos\sqrt{1-\zeta^2}\,\omega_0 t$

$(0<\zeta<1$ 欠阻尼$)$ $+\left(X_0 - X_{\infty 0} + \frac{\dot{X}_0 - \dot{X}_{\infty 0}}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_0 t} \sin\sqrt{1-\zeta^2}\,\omega_0 t$

② $\zeta=1$ 临界阻尼：对上式取 $\lim_{\zeta\to 1^-}$

③ $\zeta>1$ 过阻尼：$\sqrt{1-\zeta^2} \to \sqrt{\zeta^2-1}$；$\sin, \cos \to \sinh, \cosh$

④ $\zeta=0$ 无阻尼：$\zeta=0$ 代入，

$$X(t) = X_{\infty}(t) + \sqrt{(X_0 - X_{\infty 0})^2 + \left(\frac{\dot{X}_0 - \dot{X}_{\infty 0}}{\omega_0}\right)^2} \cos\left(\omega_0 t - \arctan\frac{\dot{X}_0 - \dot{X}_{\infty 0}}{\omega_0(X_0 - X_{\infty 0})}\right)$$

振幅不变

注：对于欠阻尼，$X(t)$ 可写作 $X_\infty(t) + Ae^{-\frac{t}{\tau}}\cos(\omega_d t - \varphi_2)$

经过Q个T，振铃衰减为	振铃
1.5Q	1%
2.2Q	0.1%

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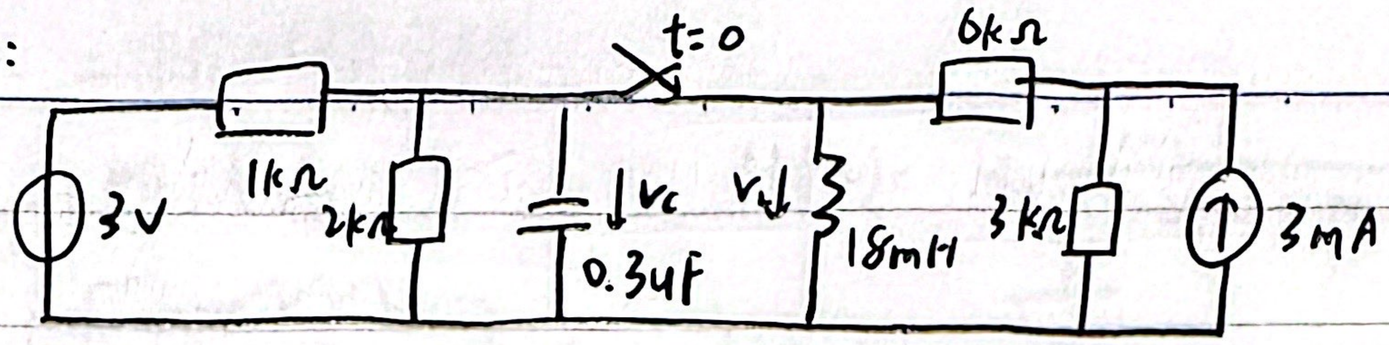
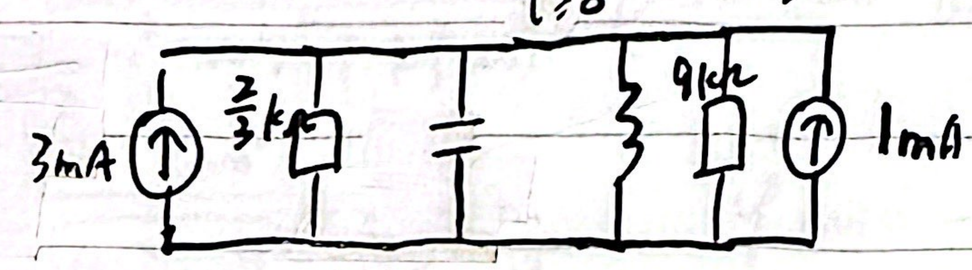
例：



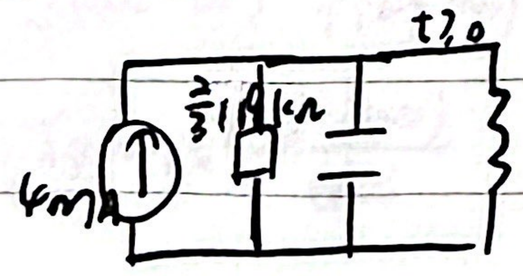
开关闭合前电路已稳定，求闭合后 $V_c(t)$, $i_L(t)$

$$V_c(t) = V_{c\infty}(t) + [V_c(0^+) - V_{c\infty}(0^+)]e^{-\zeta\omega_0 t}\cos\sqrt{1-\zeta^2}\,\omega_0 t + \left(V_c(0^+) - V_{c\infty}(0^+) + \frac{\dot{V}_c(0^+) - \dot{V}_{c\infty}(0^+)}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\sqrt{1-\zeta^2}\,\omega_0 t$$

原 (=)



(=)



① $\omega_0 = \frac{1}{\sqrt{LC}}$ ② $\zeta = \frac{1}{2Q}$ $Q = \frac{Y}{G} = \frac{\sqrt{\frac{C}{L}}}{G}$

③ $V_c(0^+) = 2V$ ④ $V_{c\infty}(t) = 0$ ⑤ $\dot{V}_c(0^+) = \frac{1}{C} i_C(0^+)$

$\dot{V}_{c\infty}(0^+) = 0$

$= \frac{1}{C}\left(i_S(0^+) - i_R(0^+) - i_L(0^+)\right)$

$= \frac{1}{C}\left(4mA - \frac{2V}{(\frac{2}{3}\|9)k\Omega} - 1mA\right)$

① $\omega_0 = \frac{1}{\sqrt{LC}}$ ② $\zeta = \frac{1}{2Q}$ $Q = \frac{Y}{G} = \frac{\sqrt{\frac{C}{L}}}{G}$

③ $i_L(0^+) = 1mA$ ④ $i_{L\infty}(t) = 4mA$

⑤ $\dot{i}_L(0^+) = \frac{1}{L} V_L(0^+) = \frac{1}{L} V_c(0^+)$ $\dot{i}_{L\infty}(0^+) = 0$

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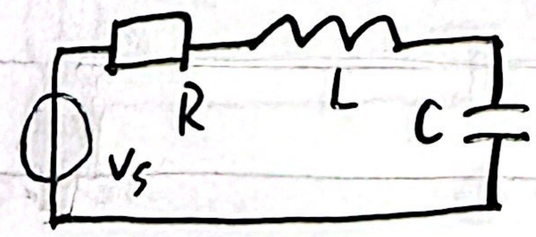
十二. 二阶时频分析

1. 二阶低通

$$H(jw) = H_0 \frac{w_0^2}{s^2 + 2\zeta w_0 s + w_0^2}$$



$$H(jw) = \frac{\frac{1}{jwC}}{\frac{1}{jwC} + jwL + R} = \frac{\frac{1}{LC}}{s^2 + s\frac{R}{L} + \frac{1}{LC}}$$

$$2\zeta w_0 \quad w_0^2$$

$$\Rightarrow H_0 = 1 \qquad w_0 = \frac{1}{\sqrt{LC}} \qquad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

$$\xrightarrow{s=jw} H(jw) = \frac{w_0^2}{\sqrt{(w_0^2 - w^2)^2 + (2\zeta w_0 w)^2}} e^{-j\arctan\frac{2\zeta w_0 w}{w_0^2 - w^2}}$$

$$A(w) = \frac{w_0^2}{\sqrt{(w_0^2 - w^2)^2 + (2\zeta w_0)^2}} \qquad \varphi(w) = -\arctan\frac{2\zeta w_0 w}{w_0^2 - w^2}$$

群延时 $\tau_g(w) = \frac{d\varphi(w)}{dw}$

... 见笔记

结论：① 幅度最大平坦：$\zeta = 0.707$

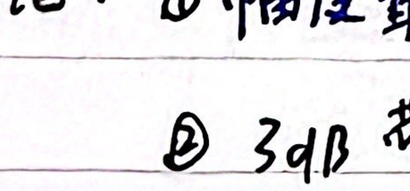
② 3dB 带宽，$BW_{3dB} = \frac{w_{3dB}}{2\pi}$

$$w_{3dB} = w_0\sqrt{-2\zeta^2 + 1 + \sqrt{(2\zeta^2 - 1)^2 + 1}} \simeq \begin{cases} w_0 & \zeta = 0.707 \\ \frac{w_0}{2\zeta} & \zeta \gg 1 \end{cases}$$

③ 群延时最大平坦：$\zeta = \frac{\sqrt{2}}{2} = 0.866$

时域特性

e.g.1 冲激响应



$$V_s(t) = \frac{V_{s0}}{w_0}\delta(t)$$

$$V_c(t) = V_{c\infty}(t) + (V_c(0^+) - V_{c\infty}(0^+))e^{-\zeta w_0 t}\cos\sqrt{1-\zeta^2}w_0 t + \left(V_c(0^+) - V_{c\infty}(0^+) + \frac{\dot{V}_c(0^+) - \dot{V}_{c\infty}(0^+)}{\zeta w_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta w_0 t}\sin\sqrt{1-\zeta^2}w_0$$

① $w_0 = \frac{1}{\sqrt{LC}}$ ② $\zeta = \frac{1}{2Q}$ $Q = \frac{w_0 L}{R}$ ③ $V_{c\infty}(t) = 0$

④ $V_c(0^+) = V_c(0^-) = 0$ 冲激 ↔ 高频 ⇒ C 短 L 开 ∴ $i_L(0^+) = \cdots = 0$, $V_L(0^+) = 0$

⑤ $\dot{V}_c(0^+) = \frac{1}{C}i_c(0^+) = \frac{1}{C}i_L(0^+) = \frac{1}{C}\left[i_L(0^-) + \frac{1}{L}\int_{0^-}^{0^+} V_L(t)dt\right] = \frac{V_{s0}}{w_0 LC}$

($i_L(0^-) = 0$, $V_L(t) = \frac{V_{s0}}{w_0}\delta(t)$)

(也可理解为：L 有界 $\therefore \frac{1}{C}\int_{0^-}^{0^+} i\,dt = 0$)

⇒ $V_c(t) = \cdots$

系数归化为 1，得 $h(t)$

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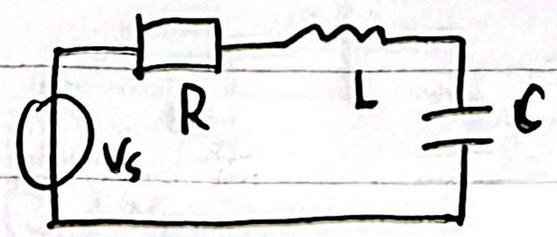
e.g.2 阶跃响应



$V_S(t) = V_{S0}\,U(t)$

① $\omega_0 = \frac{1}{\sqrt{LC}}$ ② $\zeta = \frac{1}{2Q}$ $Q = \frac{\omega_0 L}{R}$

阶跃电压不能使 i_L、V_C 突变

③ $V_C(0_+) = V_{S0}$ ④ $V_L(0^+) = 0$ ⑤ $\dot{V}_C(0^+) = \frac{1}{C} i_L(0^+) = \frac{1}{C} i_L(0^+) = 0$

$\Rightarrow V_C(t) = \cdots$ 系数归化为1, $g(t) = \cdots$

(PS: e.g.1和2的结果都要乘个U(t))

2. 二阶高通

3. 二阶带通

$$A(f) = \frac{1}{\sqrt{1+Q^2\left(\frac{f}{f_0} - \frac{f_0}{f}\right)^2}} \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$BW_{3dB} = \frac{f_0}{Q}$$

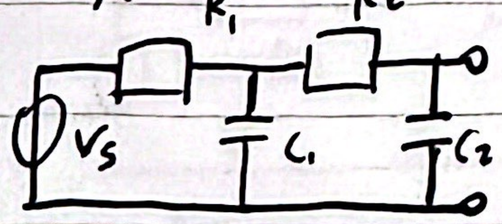
十三. 有源滤波

1. 二阶带阻

2. 有源滤波器

练(二阶节) 求 $H_S = \frac{V_o}{V_i}$

(1). 无源



$H(S) = \frac{1}{A}$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & R_1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ sC_1 & 1 \end{pmatrix}\begin{pmatrix} 1 & R_2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ sC_2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+sC_1R_1 & R_1 \\ sC_1 & 1 \end{pmatrix}\begin{pmatrix} 1+sC_2R_2 & R_2 \\ sC_2 & 1 \end{pmatrix}$$

$A = (1+sC_1R_1)(1+sC_2R_2)+sC_2$

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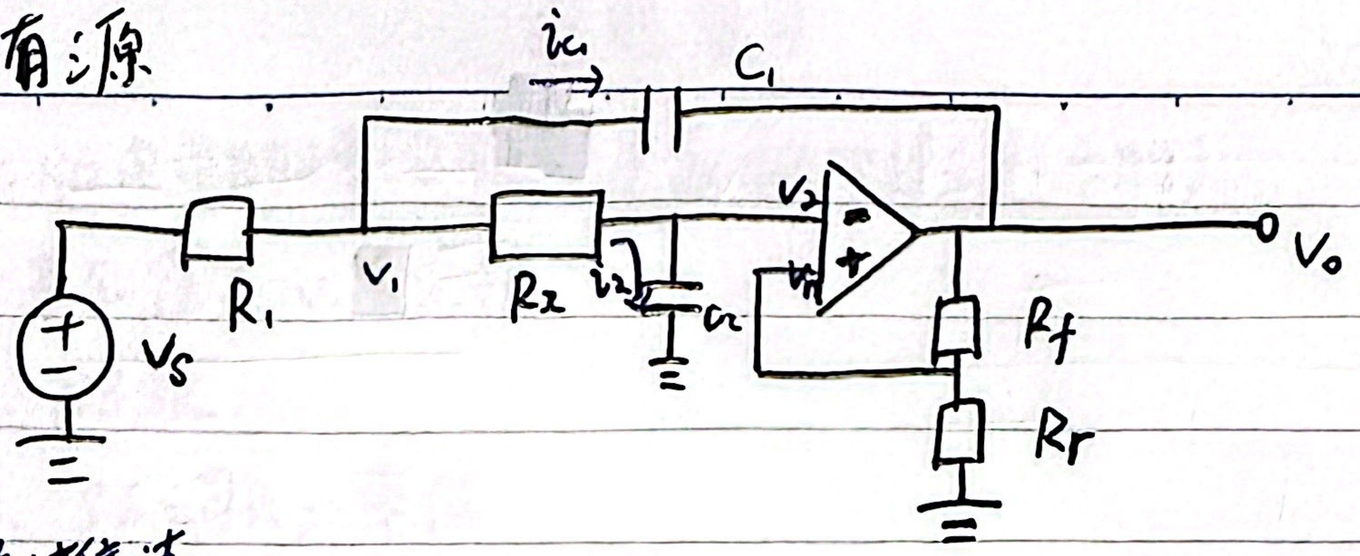
(2) 有源



倒推法

$$\dot{V}_n = \frac{R_r}{R_r+R_F}\dot{V}_o = F_n\dot{V}_o$$

$$V_2 = \dot{V}_n = F_n\dot{V}_o$$

$$\dot{V}_1 = \dot{V}_2 + i_2R_2 = \dot{V}_2 + sC_2\cdot(\dot{V}_2-0)\cdot R_2 = (1+sC_2R_2)\dot{V}_2 = (1+sC_2R_2)F_n\dot{V}_o$$

$$\dot{V}_s = \dot{V}_1 + R_1(i_{c1}+i_{c2}) = \dot{V}_1 + R_1[(\dot{V}_1-\dot{V}_o)sC_1 + sC_2\dot{V}_2]$$

$$= (1+sC_2R_2)F_n\dot{V}_o + R_1\left([F_n(1+sC_2R_2)-1]sC_1 + sC_2F_n\right)\dot{V}_o$$

$$= \dot{V}_oF_n\left(1+sC_2R_2+sC_2R_1+sC_1R_1(1+sC_2R_2) - \frac{1}{F_n}sC_1R_1\right)$$

$$\Rightarrow H(s) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{F_nC_1R_1C_2R_2}{s^2 + s\left[\frac{1}{C_1R_1}+\frac{1}{C_1R_2}-\left(\frac{1}{F_n}-1\right)\frac{1}{C_2R_2}\right] + \frac{1}{C_1C_2R_1R_2}}$$

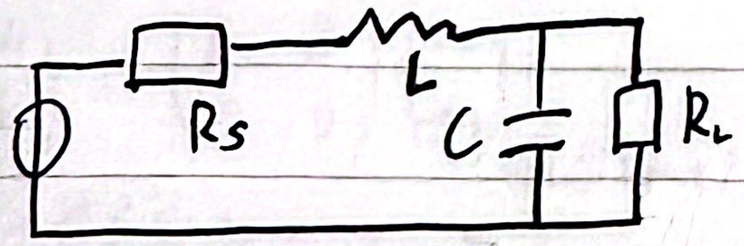
$$\omega_0 = \frac{1}{\sqrt{C_1C_2R_1R_2}} \qquad \zeta = \frac{1}{2}\left(\sqrt{\frac{C_2R_2}{C_1R_1}} + \sqrt{\frac{C_2R_1}{C_1R_2}} + \sqrt{\frac{C_2R_2}{C_1R_1}} - F_n\sqrt{\frac{C_2R_1}{C_1R_2}}\right)$$

↑ 正反馈使ζ↓，应使其＜负反馈

3. LC匹配网络

口诀：并大串小Q相等 $\quad Q = \sqrt{\frac{R_{大}}{R_{小}}-1} = Q_{并} = Q_{串}$

e.g.



$$Q = \sqrt{\frac{R_L}{R_s}-1} = \frac{\omega_0C}{G_L} = \frac{\omega_0L}{R_s}$$

计已知 R_L, R_s, ω_0，可求 C.L

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